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EPSON OPOS ADK MANUAL

APPLICATION DEVELOPMENT GUIDE

POSPrinter (TM-L60II)

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Contents

Section 1. Introduction

This manual describes the method of use and related items, as well as machine-specific precautions, when the EPSON TM-L60II Series POS Printers are used with the EPSON OPOS ADK program.

This manual applies to the following devices.

Device List

Serial	Parallel
TM-L60II	TM-L60IIP

Before reading the manual, see the following explanation about the characteristic of the TM-L60II models.

- Station: Receipt (Line thermal 180 dpi X 180 dpi)
- Label printing.

Throughout the manual, the various model names will be referred to as TM-L60II.

Compatibility mode

The compatibility mode for upward compatibility was added in OPOS Ver2.60.

For the details of the compatibility mode, please refer to “EPSON OPOS ADK MANUAL APPLICATION DEVELOPMENT GUIDE Compatibility Mode”.

Section 2.　Details on Settings

This section describes connection configurations and how to make the settings for the TM-L60II Series printers.

2.1　References of Firmware Versions

Refer to the release notes (Relnote.txt).

2.2　Settings of DIP Switches

Confirm that the following settings have been made correctly.

1)　Serial port

DIP-SW1

No.	Setting	
1	OFF	Recommended
2	OFF	Recommended
3	OFF	Fixed at OFF
4	OFF	Fixed at OFF
5	OFF	Settable
6	OFF	Settable
7	ON	Settable
8	OFF	Settable

DIP-SW2

No.	Setting	
1	OFF	Settable
2	OFF	Settable
3	OFF	Settable
4	OFF	Settable
5	ON	Fixed at ON
6	ON	Settable
7	OFF	Fixed at OFF
8	OFF	Fixed at OFF

It is possible to change the settings of DIP-SW1-1 (Processing of the data input error) and DIP-SW1-2 (Specification of the received buffer capacity), but it is recommended to leave them OFF.
Set DIP-SW1-3 (Handshake) to DTR/DSR.
Set DIP-SW1-4 (Bit length) to 8 bits.
Set DIP-SW1-5 to DIP-SW1-8 in accordance with the port information.
The described set values are the default values. For the details, refer to the product manual of the POSPrinter. Also, if these settings are changed, make sure to change the port information using the SetupPOS utility.
Set DIP-SW2-2 and DIP-SW2-3 (Specification of the print density)　to match the environment of use.

Set DIP-SW2-6 (Select paper) to match the environment of use.
Make other settings in accordance with the settings described above.

2) Parallel Port

DIP-SW1

No.	Setting	
1	OFF	Recommended
2	OFF	Recommended
3	OFF	Settable
4	OFF	Settable
5	OFF	Settable
6	ON	Fixed at ON
7	OFF	Settable
8	OFF	Fixed at OFF

It is possible to change the settings of DIP-SW1-1 (Auto line feed) and DIP-SW1-2 (Specification of the received buffer capacity), but it is recommended to leave them OFF.
Set DIP-SW1-4 and DIP-SW1-5 (Specification of the print density) of to match the environment of use.
Set DIP-SW1-7 (Select paper) to match the environment of use.
When parallel I/F is used with Windows 2000, Windows XP or Windows Vista, please set Busy Condition of DIP-SW1-3 to ON (Buffer full).
Make other settings in accordance with the settings described above.

2.3 Port Information

1) Port information when serial port is used

The port information that can be set with the SetupPOS utility is as follows.

Item	Setting range
Baud rate [bps]	2400,4800,9600,19200
Bit length [bit]	8
Parity	NONE, ODD, EVEN
Stop bit [bit]	1
Handshake	DTR/DSR
Output buffer length [byte]	32 to 1024
Output interval time [ms]	0 to 9999

The default settings are as shown in the following table.

Item	Setting range
Baud rate [bps]	9600
Bit length [bit]	8
Parity	NONE
Stop bit [bit]	1
Handshake	DTR/DSR
Input buffer length [byte]	1024
Output buffer length [byte]	1024
Output interval time [ms]	2500

2) Port information when using parallel port

The port information that can be set with the SetupPOS utility is as follows.

Item	Setting range
Output buffer length [byte]	32 to 1024
Output interval time [ms]	0 to 9999

The default settings are as shown in the following table.

Item	Setting range
Output buffer length [byte]	1024
Output interval time [ms]	2500

2.4 Device Settings

The following explanation is about the settings for TM-L60II.

2.4.1 Usable Device Specific Settings

For the TM-L60II, the following device specific settings are settable by the SetupPOS utility. For the detail, please refer to the Section 2 of “EPSON OPOS ADK MANUAL APPLICATION DEVELOPMENT GUIDE POSPrinter (TM Series)”.

Tab	Settings
General	Disable panel buttons
	Assume print complete when data output finishes
	Ignore firmware version check
	Homogenize Error Codes
	Output complete timeout
Color Bitmap	Method
	Brightness
	Primary
Status Log	ERROR
	OFFLINE
	Log file name (full path name)
	Maximum file size [KB]

Section 3. Function Details

This section describes the functions of the TM-L60II printers in details. Supplementary explanation of the parts not described in detail in the "UPOS" is also given here.

3.1 Property Set Values and Default Values

The following explanation is about the property set values and the default values.

3.1.1 Capability Set Values

The following values are the Capability set values.

Capability Name	Set Value
CapTransaction	TRUE
CapCoverSensor	TRUE
CapConcurrentRecSlp	FALSE
CapConcurrentJrnSlp	FALSE
CapConcurrentJrnRec	FALSE
CapConcurrentPageMode	FALSE
CapCharacterSet	PTR_CCS_ASCII
CapMapCharacterSet	FALSE
CapJrnUnderline	FALSE
CapJrnNearEndSensor	FALSE
CapJrnItalic	FALSE
CapJrnEmptySensor	FALSE
CapJrnDwideDhigh	FALSE
CapJrnDwide	FALSE
CapJrnDhigh	FALSE
CapJrnColor	0
CapJrnCartridgeSensor	0
CapJrnBold	FALSE
CapJrn2Color	FALSE
CapJrnPresent	FALSE
CapRecPageMode	TRUE
CapRecUnderline	TRUE
CapRecStamp	FALSE
CapRecRotate180	TRUE
CapRecRight90	TRUE
CapRecPapercut	FALSE

CapRecNearEndSensor	TRUE
CapRecMarkFeed	PTR_MF_TO_TAKEUP PTR_MF_TO_NEXT_TOF
CapRecLeft90	TRUE
CapRecItalic	FALSE
CapRecEmptySensor	TRUE
CapRecDwideDhigh	TRUE
CapRecDwide	TRUE
CapRecDhigh	TRUE
CapRecColor	PTR_COLOR_PRIMARY
CapRecCartridgeSensor	0
CapRecBold	TRUE
CapRecBitmap	TRUE
CapRecBarCode	TRUE
CapRec2Color	FALSE
CapRecPresent	TRUE
CapSlpUnderline	FALSE
CapSlpRotate180	FALSE
CapSlpRight90	FALSE
CapSlpNearEndSensor	FALSE
CapSlpLeft90	FALSE
CapSlpItalic	FALSE
CapSlpEmptySensor	FALSE
CapSlpDwideDhigh	FALSE
CapSlpDwide	FALSE
CapSlpDhigh	FALSE
CapSlpColor	0
CapSlpCartridgeSensor	0
CapSlpBothSidesPrint	FALSE
CapSlpBold	FALSE
CapSlpBitmap	FALSE
CapSlpBarCode	FALSE
CapSlp2Color	FALSE
CapSlpFullslip	FALSE
CapSlpPresent	FALSE
CapSlpPageMode	FALSE

3.1.2 List Properties

The List Properties are explained in the following.

List Property	Settings
CharacterSetList	"437,850,858,860,863,865,998"
JrnLineCharsList	""
RecLineCharsList (Label Paper) *1	"30,40"
RecLineCharsList (Normal Paper) *1	"32,42"
SlpLineCharsList	""
RecBarCodeRotationList	"0,R90, L90, 180"
RecBitmapRotationList	"0,R90, L90, 180"
SlpBarCodeRotationList	""
SlpBitmapRotationList	""
FontTypefaceList	""

*1 In selecting paper type with the DIP-SW, specified paper type (Label Paper or Normal Paper) is settable.

3.1.3 Width and Height Properties

The width and height properties are described below.

Property	Settings		
	Default Value	Maximum value [dot]	Minimum value [dot]
RecLineSpacing	30	127	24
JrnLineSpacing	X	X	X
SlpLineSpacing	X	X	X
SlpLineHeight[dot]	X		
RecLineHeight[dot]	24		
JrnLineHeight[dot]	X		
SlpLineWidth[dot]	X		
RecLineWidth[dot] (Label Paper) *1	368		
RecLineWidth[dot] (Normal Paper) *1	384		
JrnLineWidth[dot]	X		
RecSidewaysMaxLines	12 *3		
RecSidewaysMaxChars (Select Font A)	94 *4		
RecSidewaysMaxChars (Select Font B)	123 *4		
RecLinesToPaperCut	5 *2		
SlpSidewaysMaxLines	X		
SlpSidewaysMaxChars	X		
SlpMaxLines	X		

X : No settings

*1 In selecting paper type with the DIP-SW, specified paper type (Label Paper or Normal Paper) is settable.

[*2] It can be changed by the settings of the RecLineSpacing or the character height.
[*3] It can be changed by the settings of the XxxLineSpacing or the XxxLineHeight.
Also, it depends on the paper setting (Label Paper or Normal Paper).
[*4] It can be changed by the settings of the font width.

3.1.4 Common Property Strings

The Device information properties are described below.

I/F	DeviceName	DeviceDescription
S	TM-L60II	EPSON TM-L60II POS Printer
P	TM-L60IIP	EPSON TM-L60IIP POS Printer

I/F indicate the connected interface.

The following is the list of the two connecting interfaces.

S: Serial

P: Parallel

3.1.5 PageMode Print Properties

The Device information properties are described below.

- TM-L60II (Roll paper)

Property	Station [*2]		
	Journal	Receipt	Slip
PageModeArea	-	“384”, “1108”	-
PageModeDescriptor [*1]	-	BM/BC/BMR/BCR	-

- TM-L60II (Label paper)

Property	Station [*2]		
	Journal	Receipt	Slip
PageModeArea	-	“368”, “1108”	-
PageModeDescriptor [*1]	-	BM/BC/BMR/BCR	-

[*1] Following setting values are used for the PageModeDescriptor property.
BM : Bitmap printing is available.
BC : Barcode printing is available.
BMR : Rotated printing of bitmap is available.
BCR : Rotated printing of barcode is available.
[*2] If the Station’s CapRecPageMode and/or CapSlpPageMode property values are FALSE, the PageModeArea property shall have ” ” and the PageModeDescriptor property shall have ”0” respectively as a setting value.

3.2 Methods

The following explanation is about supported/unsupported Methods, and the detailed information.

Method	Supported/Unsupported	Compatibility with the PageMode printing
PrintNormal	O	O
PrintTwoNormal	X	X
PrintImmediate	O	O *2
PrintBarCode	O	O *3
PrintBitmap	O	O *4
PrintMemoryBitmap	O	O *4
CutPaper	X	X
MarkFeed	O *1	O
ChangePrintSide	X	X
ValidateData	O	O
TransactionPrint	O	O
SetLogo	O	O
SetBitmap	O	O
RotatePrint	O	X
EndRemoval	X	X
BeginRemoval	X	X
EndInsertion	X	X
BeginInsertion	X	X
ClearPrintArea	O	O
PageModePrint	O	O

O : Supported

X : Unsupported

*1 Available only for select label paper. On the other models, there is X setting.

*2 If the specified Station is ready to print, the printing data shall not be stored in the PageMode printing buffer but, instead, go straight to printing. If the Station is not ready to print, an error is returned.

*3 If other than “LEFT” is specified for the printing position of barcode, the printing shall be done, regardless of the PageModeHorizontalPosition property setting, based on the PageModePrintArea property setting in the horizontal direction.

*4 If other than “LEFT” is specified for the printing position of bitmap, the printing shall be done, regardless of the PageModeHorizontalPosition property setting, based on the PageModePrintArea property setting in the horizontal direction.

3.3 Escape Sequences

The following figure is about supported/unsupported Escape Sequences.

Escape Sequence	Value	Compatibility with the PageMode printing
#P	X	X
#fP	X	X
#sP	X	X
sL	X	X
#B	O	O
tL	O	O
bL	O	O
#R	O	O
#lF	0~9999	O
#uF Base Pitch [inch]	0~ approx. 50 cm	O
#rF Maximum [inch]	X	X
#E	0~65535	X
#fT	X	X
bC	O	O
!bC	O	O
#uC	1~2	O
iC	X	X
!iC	X	X
#rC	1	O
rvC	O	O
!rvC	O	O
#sC	X	X
#fC	X	X
tbC	X	X
!tbC	X	X
tpC	X	X
!tpC	X	X
1C	O	O
2C	O	O
3C	O	O
4C	O	O
#hC	1~8	O
#vC	1~8	O
cA	O	O [*1]
rA	O	O [*1]
lA	O	O
N	O	O

O: Supported

X : Unsupported

Numbers: Settable range

[*1] Regardless of the PageModeHorizontalPosition property setting, center or right adjust what is to be printed based on the PageModePrintArea property setting in the horizontal direction.

3.4 Printable Barcode Type

The TM-L60II models allow the following barcode types.

- Code 128
- Code 128 Parsed
- Code 93
- Codabar
- ITF
- Code 39
- JAN 13 (EAN 13)
- JAN 8 (EAN 8)
- UPC-E
- UPC-A

3.5 Power Condition Reports

The TM-L60II models do not support Power Condition Reports.

3.6 Synchronous Processing

The TM-L60II models do not use Process ID to determine output completion.

3.7 Printing Positions

The TM-L60II models support the function for setting printing position.

Function	Support
Left margin	O
Printing Position	O

O: Supported

X : Unsupported

When the left margin setting function is supported, it is possible to specify the horizontal printing position of the bitmap or barcode by dots unit.

When the printing position settings are supported, it is possible to specify the horizontal printing position of the text, bitmap, or the barcode to the left, center, or the right side of the paper.

3.8 Electronic Logo Function (NVRAM)

The TM-L60II models do not support the Electronic Logo Function.

3.9 Printable Bitmap Types and Sizes

The TM-L60II models support the following bitmap commands. For the detail, please refer to the Section 3 of “EPSON OPOS ADK MANUAL APPLICATION DEVELOPMENT GUIDE POSPrinter (TM Series)”. The allowance ranges for bitmaps are as follows.

Bitmap command type	Allowance range		
Download bitmap	x (dot)	y (dot)	xy
	1~2040	1~384	<= 98304
One-line bitmap	No setting range		

Even if meet with the limitation described above, a bitmap that extend the paper width cannot be printed.

3.10 Maintenance Counter

The TM-L60II models do not support the Maintenance Counter.

3.11 Automatic Recovery Function

The TM-L60II models do not have a function for automatic recovery when the power is turned on again after an interruption of power.

3.12 MarkFeed Function

The TM-L60II models support the MarkFeed function. When executing the PTR_MF_TAKEUP, the power status becomes OFFLINE. The printer will remain OFFLINE until the [FEED] button is pressed.

3.13 Label Receipt DirectIO

The TM-L60II models support the following DirectIO commands. For usage information, please refer to the Section 4 of "EPSON OPOS ADK MANUAL APPLICATION DEVELOPMENT GUIDE POSPrinter (TM Series)".

PTR_DI_LABEL_REMOVE
PTR_DI_LABEL_SET_PRINT_MODE
PTR_DI_LABEL_SET_COUNT_MODE
PTR_DI_LABEL_PRINT_COUNT
PTR_DI_LABEL_SET_COUNT_VALUE

3.14 Label receipt Usage

- When using label receipt paper with TM-L60II (Serial), set DIP-SW2-6 to ON, with TM-L60IIP (Parallel), set DIP-SW1-7 to ON. If the label receipt paper DIP-SW is set, be sure label receipt paper is in the printer.
- When using label receipt paper, the Escape Sequence ESC|#IF feeds the receipt the specified length, or to the head of the next label.
- After executing MarkFeed (PTR_MF_TO_TAKEUP), the power state of the printer becomes OFFLINE. The printer will remain OFFLINE until the [FEED] button is pressed. After the [FEED] button is pressed operation continues.

Section 4. Warnings

This section describes precautions in use of the TM-L60II.

- The width limitation on 90-degree rotated printing to the right is up to 831 dots.
- Horizontal printing range differs in selecting Label Paper or Normal Paper. For this reason, values of the RecLineWidth, RecLineChars, and the RecSideWaysMaxLines properties depend on the selected paper type. When using Label Paper, values of those properties are to be modified to the specific values for Label Paper after executing the ClaimDevice method.
- It takes time until the printing start when the printing data which contain bitmap data. However, it is the influence of the firmware specification.